

Name: _____

Date: _____

RATIO AND PROPORTION, (DIVIDING QUANTITIES IN A GIVEN RATIO)

LO: I can divide a quantity into two or more parts using a given ratio.

Dividing in a ratio

Dividing in a ratio means splitting a total into parts whose sizes follow the given **ratio**.

Steps: Add the ratio parts to get the total number of shares. Divide the quantity by this total to get one share. Multiply each part of the ratio by the value of one share.

Quick examples

●	£50 in ratio 2 : 3 = £20 and £30	5 parts, £10 each
●	120 in ratio 1 : 2 : 3 = 20, 40, 60	6 parts, 20 each
●	Check: shares must add to original total	$20 + 30 = 50$
●	£50 in ratio 2 : 3 = £25 and £25	that is halving, not using the ratio

Key idea

Total shares = sum of ratio parts. Value of one share = quantity divided by total shares.

$$\text{one share} = \text{quantity} \div (a + b)$$

Then multiply: a shares and b shares

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - two-part ratio

Divide £80 in the ratio 3 : 5

Total parts = $3 + 5 = 8$
 One part = $£80 \div 8 = £10$
 3 parts = £30, 5 parts = £50

£30 and £50

Check: $£30 + £50 = £80$.

Example 2 - three-part ratio

Divide 180 cm in the ratio 2 : 3 : 4

Total parts = $2 + 3 + 4 = 9$
 One part = $180 \div 9 = 20$ cm
 Shares: 40 cm, 60 cm, 80 cm

40 cm, 60 cm, 80 cm

Check: $40 + 60 + 80 = 180$.

Example 3 - working backwards

Two amounts in ratio 3 : 7 differ by £20. Find each amount.

Difference in parts = $7 - 3 = 4$ parts
 4 parts = £20, so 1 part = £5
 3 parts = £15, 7 parts = £35

£15 and £35

Use the difference when the total is not given.

Example 4 - problem in context

Mortar is cement and sand in ratio 1 : 5. A builder needs 42 kg of mortar. Find the amounts.

Total parts = $1 + 5 = 6$
 One part = $42 \div 6 = 7$ kg
 Cement = 7 kg, Sand = 35 kg

7 kg cement and 35 kg sand

Real-world ratios work exactly the same way.

YOUR TURN – PRACTICE (deliberate practice)

1. Two-part ratios

Divide each quantity in the given ratio.

- a) £40 in ratio 1 : 3
- b) 72 in ratio 5 : 4
- c) £120 in ratio 7 : 5
- d) 200 g in ratio 3 : 2

2. Three-part ratios

Divide in the ratio given.

- a) 90 in ratio 1 : 2 : 3
- b) £200 in ratio 2 : 3 : 5
- c) 360 in ratio 4 : 5 : 3
- d) £480 in ratio 1 : 3 : 4

3. Using the difference

Use the difference between shares to solve.

- a) Ratio 2 : 5. Difference is 21. Find each part.
- b) Ratio 3 : 8. Difference is 30. Find each part.
- c) Ratio 1 : 4. Larger part is 60 more. Find each.
- d) Ratio 5 : 9. Difference is 28. Find the total.

4. In context

Solve each word problem.

- a) Cordial : water = 1 : 4. Make 2 litres of drink. How much cordial?
- b) Boys : girls = 3 : 5. There are 32 students. How many boys?
- c) Cement : sand : gravel = 1 : 2 : 4. Need 35 kg total. Find each.
- d) Savings : spending = 2 : 7. Earn £270 per week. How much saved?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To divide £60 in ratio 1 : 2, find total parts (3), then one part (£20) ☐

Correct answer: _____

Reason: _____

b) Dividing £60 in ratio 1 : 2 gives £30 and £30 ☐

Correct answer: _____

Reason: _____

c) The shares must always add up to the original quantity ☐

Correct answer: _____

Reason: _____

d) Ratio 3 : 5 means the first share is always £3 ☐

Correct answer: _____

Reason: _____

e) If ratio is 2 : 3 and difference is 10, one part = 10 ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Ali, Ben and Cara share £540 in the ratio 2 : 3 : 4. Cara gives $\frac{1}{3}$ of her share to Ali. How much does each person now have?

Expression: _____

Simplified: _____

b) A 750 ml bottle of squash makes drinks by mixing with water in ratio 1 : 4. How many 250 ml glasses of drink can be made from one bottle?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

